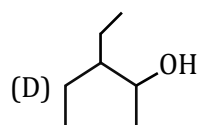
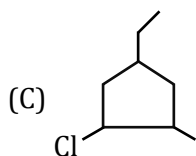
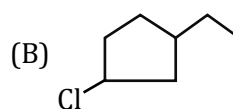
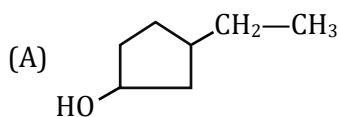
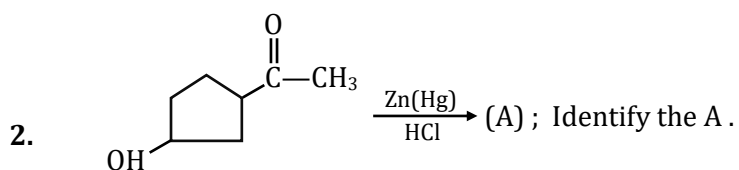
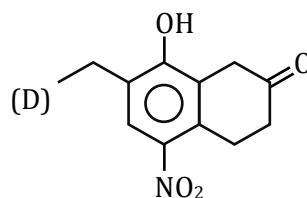
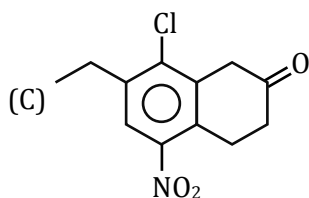
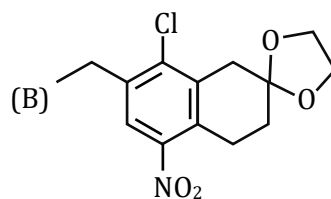
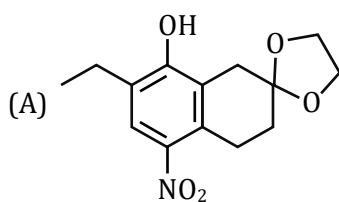
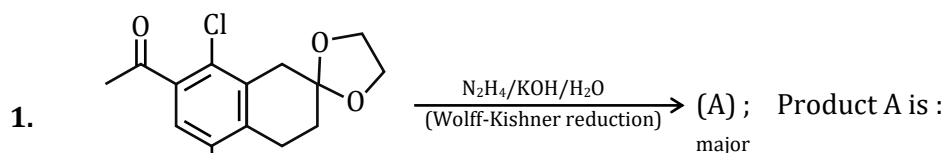
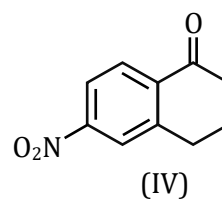
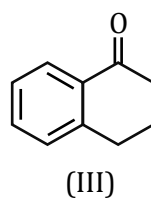
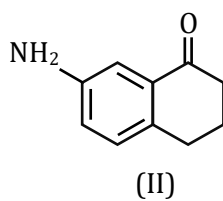
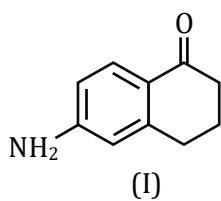




## Carbonyl Compounds : RACE-01

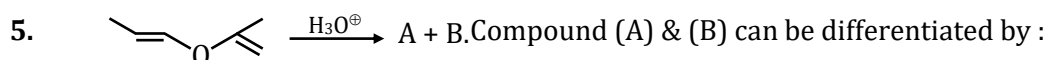
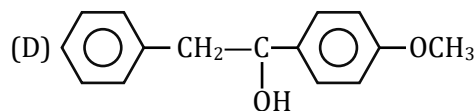
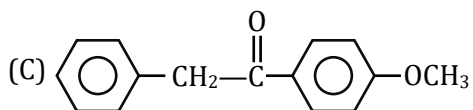
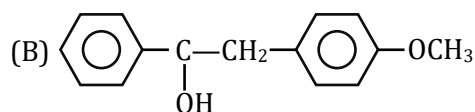
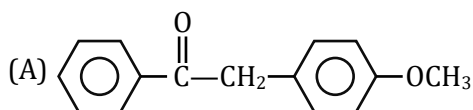
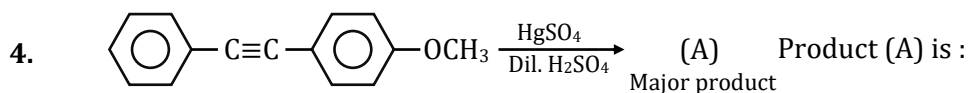


3. Increasing order of equilibrium constants for the formation of a hydrate:



(A) IV < III < II < I  
(C) I < II < III < IV

(B) IV < III < I < II  
(D) II < III < I < IV

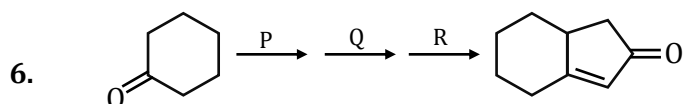


(A) 2-4-DNP

(B) Fehling solution

(C) Lucas reagent

(D) NaHSO<sub>3</sub>



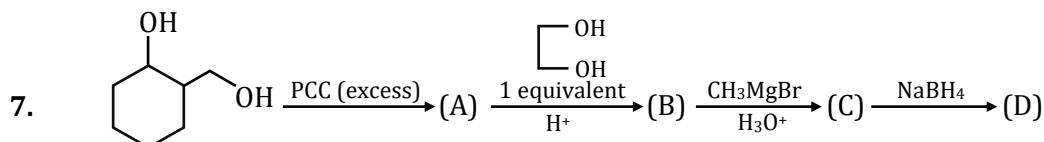
Reagents to carry out above conversion, P, Q, R respectively are :

(A) H<sub>2</sub>C=CH—CH<sub>2</sub>—Br, (HO<sup>⊖</sup>), [HO<sup>⊖</sup>, Δ], Wacker-process

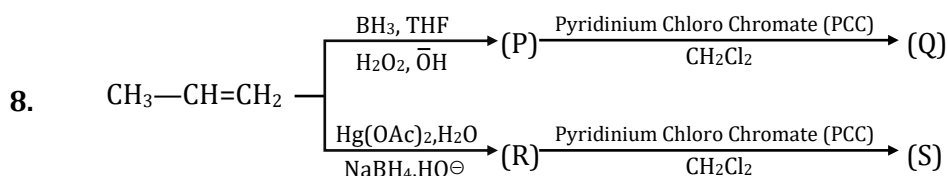
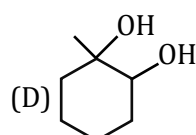
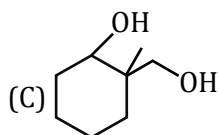
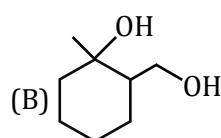
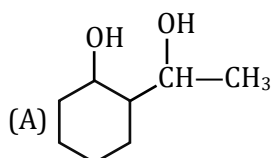
(B) H<sub>2</sub>C=CH—CH<sub>2</sub>—Br(HO<sup>⊖</sup>), Wacker-process, HO<sup>⊖</sup> Δ

(C) Wacker-process, H<sub>2</sub>C=CH—CH<sub>2</sub>—Br(HO<sup>⊖</sup>), HO<sup>⊖</sup>(Δ)

(D) Wacker-process, HO<sup>⊖</sup>(Δ), H<sub>2</sub>C=CH—CH<sub>2</sub>—Br(HO<sup>⊖</sup>)



Product (D) will be:



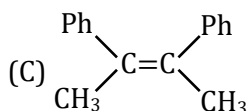
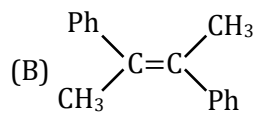
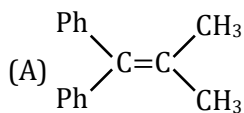
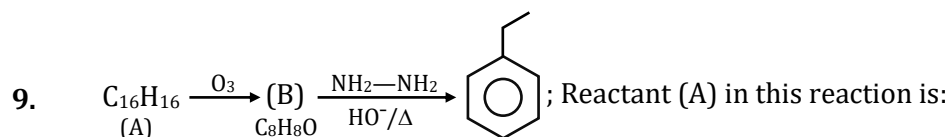
Relationship between products (Q) and (S) is:

(A) Positional isomer

(B) Chain isomer

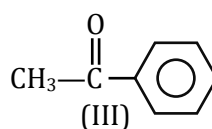
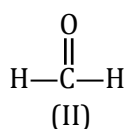
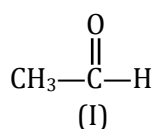
(C) Stereoisomer

(D) Functional isomer



(D) both (B) and (C)

10. Correct order of reactivity of following compounds towards Grignard reagent ?



(A) I > II > III

(B) II > I > III

(C) II > III > I

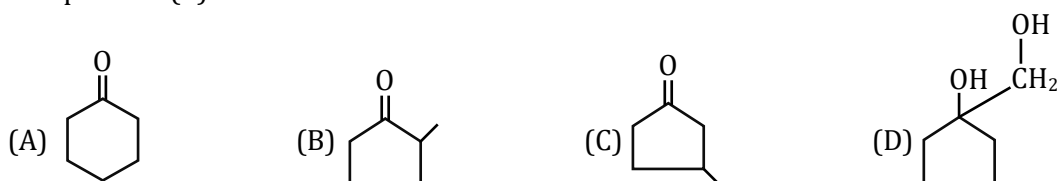
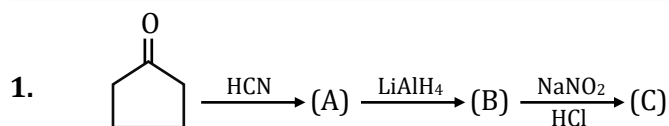
(D) I > III > II

## ANSWER KEY

| Question | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|----------|---|---|---|---|---|---|---|---|---|----|
| Answer   | A | B | C | C | B | B | B | D | D | B  |



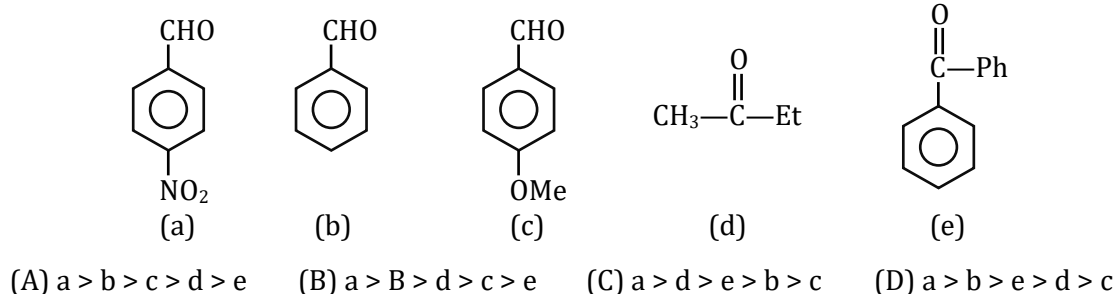
Carbonyl Compounds : RACE-02



2. Compound (X)  $C_4H_8O$ , which reacts with 2, 4-DNP derivative and gives negative haloform test is :



3. The correct order of rate of reaction toward nucleophilic addition reaction:



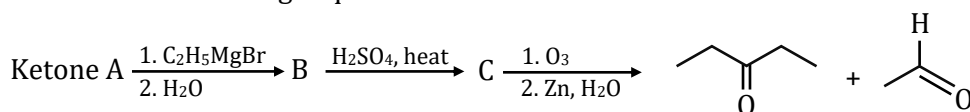
4. The structure would be best classified as a (an) :

(A) Acetal (B) Hemiacetal (C) Hydrate (D) Cyanohydrin

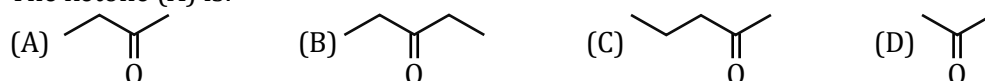
5. The reaction of  $C_6H_5CH=CHCHO$  with  $LiAlH_4$  gives:

(A)  $C_6H_5CH_2CH_2CH_2OH$  (B)  $C_6H_5CH=CHCH_2OH$   
(C)  $C_6H_5CH_2CH_2CHO$  (D)  $C_6H_5CH_2CHOHCH_3$

6. Consider the following sequence of reactions.



The ketone (A) is:



7. The presence of unsaturation in organic compounds can be tested with :  
 (A) Schiff's reagent (B) Tollens' reagent (C) Fehling's reagent (D) Baeyer's reagent

8. Which of the following gives iodoform test ?  
 (A)  $\text{CH}_3\text{CH}_2\text{OH}$  (B)  $\text{C}_2\text{H}_5\text{CHO}$  (C)  $(\text{CH}_2\text{OH})_2$  (D) None of these

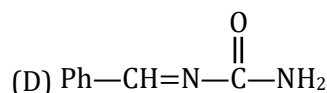
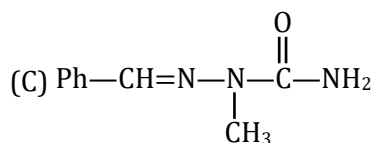
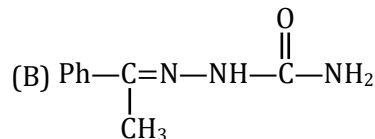
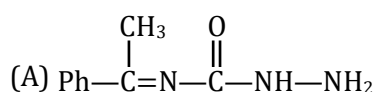
9. 
$$\begin{array}{c} \text{CHO} \\ | \\ \text{H} - \text{C} - \text{OH} \\ | \\ \text{CH}_2 - \text{OH} \end{array} \xrightarrow[\text{(ii) H}^+]{\text{(i) KCN}} \text{Product ; Product obtained in the reaction is :}$$

D-(+)-Glyceraldehyde

- (A) Diastereomer (B) Racemic  
 (C) Meso (D) Optically Pure enantiomer

10. 
$$\text{Ph} - \overset{\text{OH}}{\underset{|}{\text{CH}}} - \text{CH}_3 \xrightarrow{\text{PCC}} (\text{A}) \xrightarrow{\text{NH}_2 - \text{NH} - \overset{\text{O}}{\parallel}{\text{C}} - \text{NH}_2} (\text{B})$$

Product (B) is:



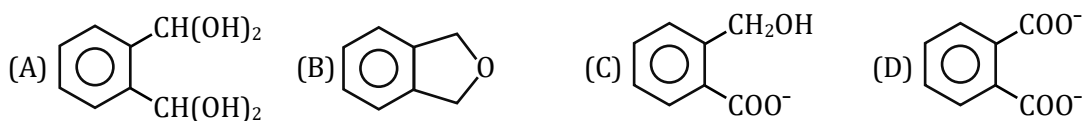
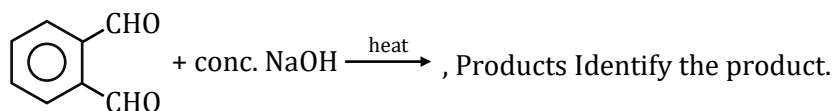
### ANSWER KEY

| Question | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|----------|---|---|---|---|---|---|---|---|---|----|
| Answer   | A | B | A | B | A | B | D | A | A | B  |



Carbonyl Compounds : RACE-03

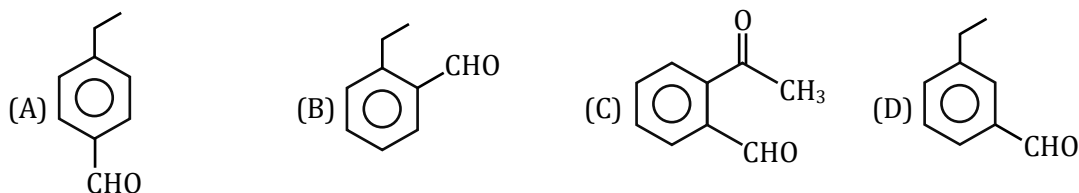
1. The reaction,



2.  $\text{H}-\overset{\text{O}}{\parallel}{\text{C}}-\text{D} \xrightarrow[(50\%)]{\text{HO}^-}$  ; Product of this Cannizaro reaction is:



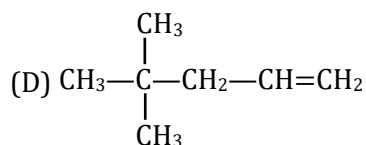
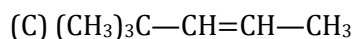
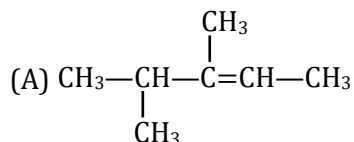
3. An organic compound with the molecular formula  $\text{C}_9\text{H}_{10}\text{O}$  forms a 2, 4-DNP derivative, reduces Tollen's reagent and undergoes Cannizaro reaction, on vigorous oxidation it gives 1, 2-benzenedicarboxylic acid. Structure of organic compound is:

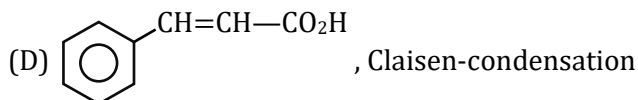
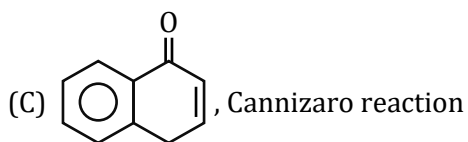
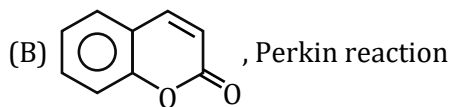
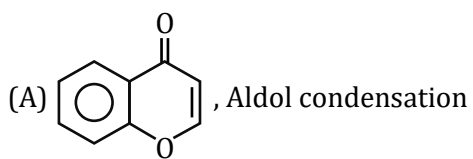
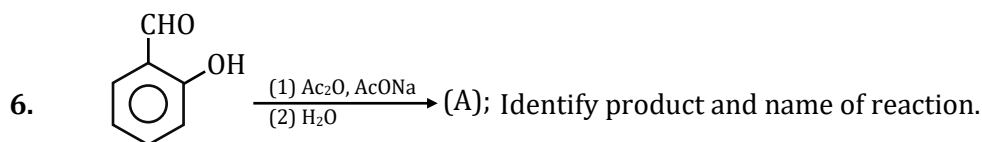
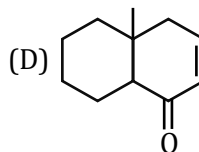
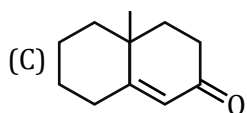
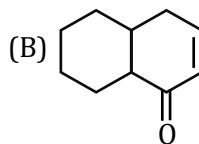
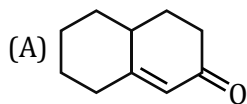
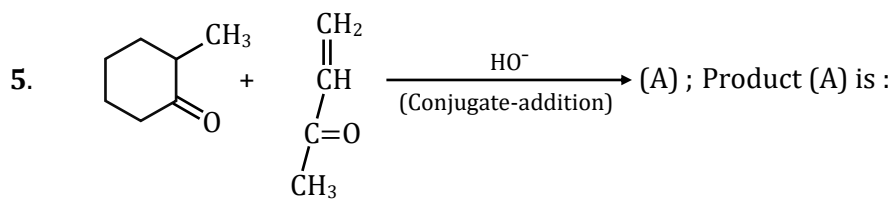


4. (A)  $\xrightarrow[\text{Zn/AcOH}]{\text{O}_3}$  (B) + (C)  
 $\text{C}_7\text{H}_{14}$

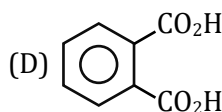
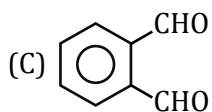
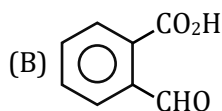
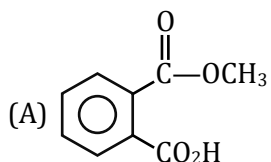
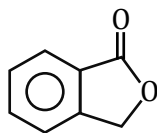
Compound (A) exist in geometrical isomers and (B) gives Cannizaro reaction.

(A) will be:

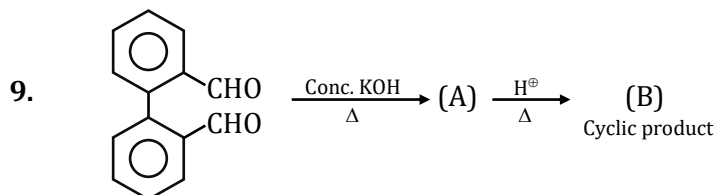
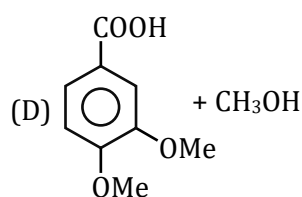
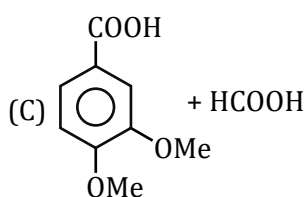
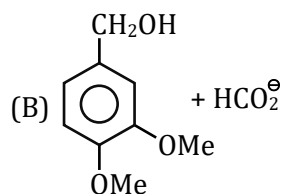
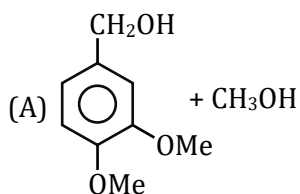
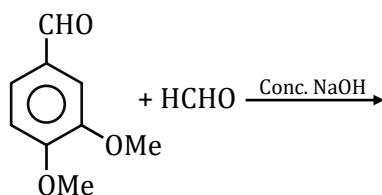




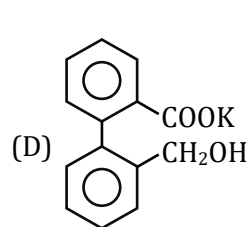
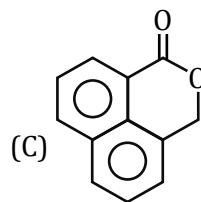
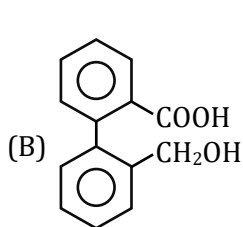
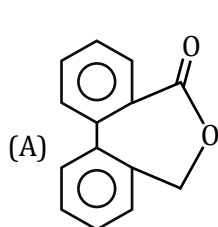
7. Which of the following reactant on reaction with conc. NaOH followed by acidification gives the following lactone as the product ?



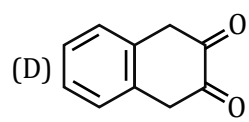
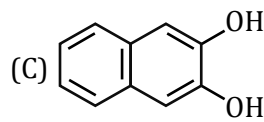
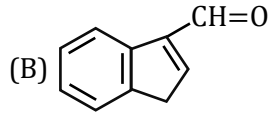
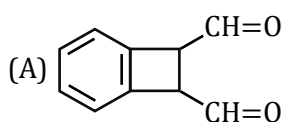
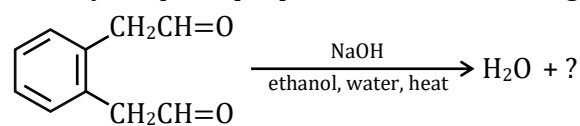
8. The following reaction gives :



Structure of (B) is:



10. Identify the principal product of the following reaction?



### ANSWER KEY

| Question | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|----------|---|---|---|---|---|---|---|---|---|----|
| Answer   | C | C | B | C | C | B | C | B | A | B  |